

Md Abdur Rahaman

Dhaka, Bangladesh — +880 1537517376 — abdur.rahaman03155@gmail.com
[LinkedIn](#) — [GitHub](#) — [Kaggle](#)

Research Summary

Computer Science and Engineering graduate with a strong academic background and research interests in Computer Vision, Deep Learning, Medical Image Analysis, and Environmental Object Detection. Experienced in developing attention-based architectures (RepViT, Transformers) for melanoma and brain tumor classification. First author of a submitted manuscript on melanoma classification using frequency-domain and attention-based deep learning. Seeking a PhD position to advance research in medical computer vision and multimodal learning.

Education

Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh
B.Sc. in Computer Science & Engineering, CGPA: 3.61 / 4.00 2022 – 2026

Relevant Coursework: Computer Vision, Deep Learning, Machine Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms.

Research Experience

Melanoma Classification Using Deep Learning *First Author*
Research Project

- **Method:** Developed a DCT-based frequency-channel attention fusion approach using RepViT for melanoma classification.
- **Results:** Achieved competitive performance on the Melanoma skin Cancer Dataset with 10000 images using fewer parameters.
- **Status:** Manuscript submitted to ICRPSET, Paper ID: 185.

Multi-Class Brain Tumor Identification from MRI *Second Author*
Research Project

- **Method:** Applied EfficientNetB1 transfer learning for four-class brain tumor classification.
- **Results:** Achieved **98.00% accuracy** and **98.00% macro F1-score** on MRI images.
- **Status:** Submitted to ICCIT 2026, Paper ID: 917.

Diabetic Retinopathy Grading Using Hybrid CNN–Transformer *Research Project*

- **Method:** Developed an EfficientNet-B2–DeiT hybrid with adaptive fusion and imbalance-aware training for five-stage grading on APTOS and DDR datasets.

Eco-YOLO: Marine Debris Detection with PG-EPE and YOLO11m *Research Project*

- **Method:** Developed Eco-YOLO, a 40-class marine debris detector combining a novel PG-EPE enhancement pipeline with YOLO11m on the SeaClear dataset.
- **Results:** Achieved **87.5% mAP@0.5**, **65.9% mAP@0.5:0.95**, **89.4% precision**, and **80.7% recall** with 20.08M parameters, 68.5 GFLOPs, and 57.6 FPS on a Tesla T4.

Manuscripts Under Review

Md Abdur Rahaman (first author). “MeLALight: Melanoma Classification Using Multi-Stage DCT-Based Frequency-Channel Attention Fusion on RepViT.” *Submitted to ICRPSET*, 2026.

Md Abdur Rahaman (co-author). “Adaptive Transfer Learning with EfficientNetB1 for Multi-Class Brain Tumor Identification in MRI Images.” *Submitted to ICCIT*, 2026.

Selected Projects

Opportunity Finder — RAG-Based Scholarship Matching

Developed a Retrieval-Augmented Generation (RAG) application using LangChain to recommend Erasmus Mundus programs. Worked with web-scraped information, Hugging Face embeddings, FAISS vector search, and LLM-based reasoning.

Computer Vision with OpenCV

Implemented computer vision and image-processing experiments using Python and OpenCV, including webcam-based color detection, HSV conversion, color masking, and bounding-box detection.

Achievements

NeurIPS Open Polymer Prediction 2025

Awarded a **Silver Medal**; ranked **112th out of 2,240 teams** for machine-learning-based polymer property prediction.

Technical Skills

Programming: Python, C, C++

AI / Machine Learning: PyTorch, Scikit-learn, Hugging Face, LangChain, RAG, CNN, RepViT, Transformers, Attention Mechanisms

Computer Vision: OpenCV, Image Classification, Medical Image Analysis

Tools: Git, GitHub, Linux/Ubuntu, VS Code, Kaggle, Streamlit, LaTeX, FAISS

Teaching Experience

Private Tutor

SSC & HSC Students

- Provided academic tutoring, conceptual guidance, problem solving, and examination preparation for SSC and HSC students.

References

Farjana Parvin

Assistant Professor, Department of CSE, RUET. Email: farjana@cse.ruet.ac.bd

Md. Mazharul Islam

Lecturer, Department of CSE, RUET. Email: mazharul.islam@cse.ruet.ac.bd

Barshon Sen

Assistant Professor, Department of CSE, RUET. Email: barshon.sen@cse.ruet.ac.bd